

Viral Mutation Classification using DNABERT-2 and LSTM

**Project report submitted in partial fulfillment of the requirements
for the award of the degree
BACHELOR OF TECHNOLOGY
in
COMPUTER SCIENCE& ENGINEERING (DATA SCIENCE)
Submitted by**

D. Vamsi Krishna (Y22CD28)

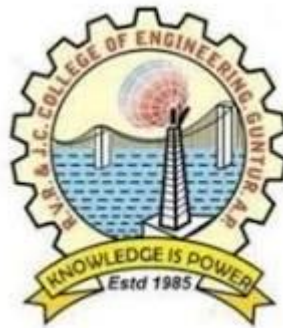
G. Samba Siva Rao (Y22CD039)

D. Ajay Kumar Reddy (Y22CD029)

Under the Guidance of

Mrs. K. Aravinda

Assistant Professor



R.V.R & J.C. COLLEGE OF ENGINEERING (AUTONOMOUS)

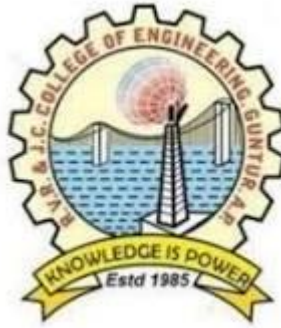
Approved by AICTE- New Delhi, Accredited by NAAC A+ Grade

Affiliated to Acharya Nagarjuna University,

Chowdavaram, Guntur - 19, Andhra Pradesh, India.

MAY 2026

R.V.R & J.C. COLLEGE OF ENGINEERING (AUTONOMOUS)
DEPARTMENT OF
COMPUTER SCIENCE & ENGINEERING (DATA SCIENCE)



CERTIFICATE

This is to certify that the project work titled “**Viral Mutation Classification using DNABERT-2 and LSTM**” is the work done by **D. Vamsi Krishna (Y22CD028)**, **G. Samba Siva Rao (Y22CD039)** and **D. Ajay Kumar Reddy (Y22CD029)** under my supervision and submitted in partial fulfillment of the requirements for the award of the degree, B.Tech. in Computer Science and Engineering (Data Science), during the Academic Year 2025-2026.

Mrs. K. Aravinda
Assistant Professor
Project Guide

Dr. M.V.P.Chandra Sekhara Rao
Professor & Head

External Examiner

DECLARATION

We **D. Vamsi Krishna (Y22CD028)**, **G. Samba Siva Rao (Y22CD039)** and **D. Ajay Kumar Reddy (Y22CD029)** hereby declare that the project report titled “**Viral Mutation Classification using DNABERT-2 and LSTM**” under the guidance of **Mrs. K. Aravinda** is submitted in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Computer Science and Engineering (Data Science). This is a record of bonafide work carried out by us and the results embodied in this project have not been reproduced or copied from any source. The results embodied in this project have not been submitted to any other university for the award of any other degree.

D. Vamsi Krishna(Y22CD028)

G. Samba Siva Rao (Y22CD039)

D. Ajay Kumar Reddy (Y22CD029)

Place: Guntur

Date:

ACKNOWLEDGMENT

The successful completion of any task would be incomplete without proper suggestion, guidance, and environment. Combination of these three factors acts like a backbone to our project work “**Viral Mutation Classification using DNABERT-2 and LSTM**”.

We are profoundly pleased to express our deep sense of gratitude and respect towards the management of the **R.V.R. & J.C. College of Engineering** for providing the resources to complete the project.

We are very much thankful to **Dr. Kolla Srinivas**, Principal, **R.V.R. & J.C. College of Engineering** for allowing us to deliver the project successfully.

We are greatly indebted to **Dr. M.V.P. Chandra Sekhara Rao**, Professor, & Head of the department, Computer Science and Engineering (Data Science), **R.V.R. & J.C. College of Engineering**, providing the laboratory facilities fully as and when required and for giving us the opportunity to carry the project work in the college.

We are also thankful to our Project Coordinator **Dr. Riaz Shaik**, Associate Professor who helped us in each step of our Project.

We are very much thankful to our Project guide **Mrs. K. Aravinda**, Assistant Professor who has inspired us to take this project and gave us valuable guidance to complete the project successfully.

We would be thankful to **all the staff** of the Department of Computer Science and Engineering (Data Science) for their valuable suggestions, guidance and constructive ideas in each and every step, which was indeed of great help towards the successful completion of our project.

D. Vamsi Krishna(Y22CD028)

G. Samba Siva Rao(Y22CD039)

D. Ajay Kumar Reddy(Y22CD029)

ABSTRACT

Recent advancements in Natural Language Processing (NLP), particularly in transfer learning, have opened new possibilities for genomic sequence analysis. This study evaluates the comparative performance of DNABERT and a hybrid DNABERT-LSTM model for mutation classification in viral genomes using datasets from the NCBI Viruses database. While DNABERT leverages k-mer tokenization and transformer-based attention mechanisms to capture rich contextual features, the LSTM component enhances the model by learning long-range dependencies within DNA sequences. The models are assessed using evaluation metrics such as accuracy, F1-score, and generalization capability, demonstrating that NLP-based deep learning approaches outperform traditional methods in mutation classification tasks. The integrated DNABERT-LSTM model achieves improved predictive performance and provides a robust and scalable framework for genomic analysis. Beyond performance metrics, this study highlights the practical significance of applying NLP techniques to biological data, enabling better understanding of viral mutation patterns and evolution. The proposed hybrid model shows strong adaptability across diverse viral datasets, making it a valuable tool for real-world applications such as mutation tracking and disease monitoring. By bridging artificial intelligence with genomics, this work contributes to the development of advanced analytical tools that support public health preparedness and precision medicine, offering a promising direction for future research in bioinformatics and computational biology.

TABLE OF CONTENTS

TITLE	PAGE No.
ABSTRACT	I
TABLE OF CONTENTS	II
LIST OF FIGURES	IV
LIST OF TABLES	V
CHAPTER 1: INTRODUCTION	
1.1. Introduction	1
1.2. Problem Statement	3
1.3. Objectives of the Study	4
CHAPTER 2: LITERATURE SURVEY	
2.1. Literature Survey	6
CHAPTER 3: SYSTEM ANALYSIS & FEASIBILITY STUDY	
3.1. Existing System	10
3.1.1. Limitations of Existing System	11
3.2. Proposed System	12
3.2.1. Advantages of Proposed System	13
3.3. Dataset	14
3.4. Data Preprocessing	17
3.4.1. Data Cleaning	17
3.4.2. Feature Understanding and Transformation	18
3.4.3. Feature Scaling	18
3.4.4. Handling Class Imbalance	19
3.4.5. Train-Test Split	20
3.5. Libraries Utilized	21
3.6. Methodology	22
3.6.1. Auto-Tokenizer	24
3.6.2. Positional Encoding	25
3.7. BERT Architecture	27
3.8. Impact of DNABERT	29
3.9. Long Short-Term Memory	31
3.10. Model Validation	33
3.11. Evaluation Metrics	34
3.12. Feasibility Study	35
3.12.1. Economic Feasibility	35
3.12.2. Operational Feasibility	35
3.12.3. Technical Feasibility	36

CHAPTER 4: SYSTEM REQUIREMENTS	
4.1. Functional Requirements	38
4.2. Non-Functional Requirements	38
4.2.1. Software Requirements	39
4.2.2. Hardware Requirements	40
CHAPTER 5: DESIGN	
5.1. System Design	42
5.2. Data Flow of System	43
CHAPTER 6: IMPLEMENTATION	
6.1. Implementation Overview	45
CHAPTER 7: RESULTS	
7.1. Result Flow	49
7.2. Training Loss Analysis	49
7.3. Training Accuracy Analysis	51
7.4. Comparison of Models	52
7.5. Results	53
CHAPTER 8: SOCIAL IMPACT	
8.1. Social Impact	54
CHAPTER 9: CONCLUSION & FUTURE WORK	
9.1 Conclusion	56
9.2 Future Work	57
CHAPTER 10: BIBLIOGRAPHY	

LIST OF FIGURES

Figure No.	Figure Description	Page No.
3.1	Methodology Overview	23
3.2	BERT Architecture	28
3.3	LSTM Network	32
5.1	System Architecture	42
7.1	Training Loss Analysis	50
7.2	Training Accuracy Analysis	51
7.3	Comparison of Models	52

LIST OF TABLES

Table No.	Table Description	Page No.
2.1	Summary of Literature Survey Approaches	8
3.1	Dataset Description	16
4.1	Software Requirements	39
4.2	Hardware Requirements	40
7.1	Experimental Results	53

Chapter 1

Introduction

1. INTRODUCTION

1.1 Introduction

Natural Language Processing (NLP) models are increasingly applied in genomic mutation detection and classification, highlighting their potential to transform genomic analysis [1], [6], [9]. Rapid advancements in sequencing technologies have resulted in the generation of massive volumes of genomic data, creating a demand for efficient, scalable, and intelligent computational approaches. Traditional bioinformatics methods, which rely heavily on alignment-based techniques, often struggle with large datasets and complex mutation patterns [8], [22]. As a result, deep learning-based approaches, particularly those inspired by NLP, have gained significant attention in recent years [1], [2].

NLP has emerged as a powerful paradigm in genomics by treating DNA sequences as a form of biological language, where nucleotides act as characters and k-mer subsequences represent words [6], [20]. This analogy allows the application of advanced language models to uncover hidden patterns within genomic data. Transformer-based architectures, such as DNABERT-2, leverage k-mer tokenization and bidirectional self-attention mechanisms to capture contextual relationships between nucleotides [10], [14], [18]. These models can learn rich representations of DNA sequences, enabling the identification of mutation patterns, regulatory elements, and functional regions without the need for manual feature engineering.

Despite their effectiveness in capturing contextual information, transformer-based models may not fully represent long-range sequential dependencies present in genomic sequences. In biological systems, interactions between distant nucleotide positions can significantly influence gene expression and mutation behavior [2], [10]. To address this limitation, Long Short-Term Memory (LSTM) networks are integrated with DNABERT-2 to enhance sequential modeling. LSTM networks are specifically designed to process sequential data and retain long-term dependencies through memory cells and gating mechanisms, including input, forget, and output gates. This enables the model to preserve important information over extended sequence lengths while filtering out irrelevant noise.

The hybrid combination of DNABERT-2 and LSTM provides a comprehensive framework that captures both contextual and sequential features of genomic data. DNABERT-2 extracts high-level contextual embeddings using attention mechanisms, while the LSTM layer refines these embeddings by learning temporal relationships across the sequence. This synergy improves the model's ability to detect subtle mutation patterns and enhances classification accuracy, especially in complex genomic datasets [4], [16].

The proposed approach has been evaluated using datasets from the NCBI Viruses database, which includes viral genomes such as Human Virus, Influenza Virus, and SARS-CoV-2. Accurate mutation classification in these datasets is crucial for understanding viral evolution, identifying emerging variants, and supporting public health initiatives. Compared to traditional methods, the hybrid model demonstrates improved performance, scalability, and adaptability for large-scale genomic analysis.

Genomic sequence classification plays a vital role in several real-world applications. In clinical diagnostics, it aids in identifying genetic disorders, predicting disease susceptibility, and guiding personalized treatment strategies. In epidemiological surveillance, mutation detection helps track the spread and evolution of infectious diseases, enabling timely intervention and control measures [22], [23]. Furthermore, it contributes to drug discovery and resistance analysis by identifying mutation patterns associated with treatment response and therapeutic effectiveness.

Overall, the integration of NLP-based deep learning techniques with genomic analysis represents a significant advancement in the field of bioinformatics. The proposed DNABERT-2 and LSTM hybrid framework provides a robust, scalable, and accurate solution for mutation detection and classification. This approach not only enhances the understanding of genomic data but also supports future developments in precision medicine, disease prediction, and large-scale biological data analysis.

.

1.2 Problem Statement

Traditional approaches for mutation classification, such as alignment-based algorithms, are limited in scope as they struggle to handle the vastness and complexity of genomic data [8], [22]. These methods rely heavily on reference genomes and often fail to accurately detect subtle mutations, especially in cases of low-quality data or newly emerging viral strains. Furthermore, aligning new viral sequences to a fixed reference genome can lead to inaccurate results, particularly when rare or previously unknown mutations are present. This limitation reduces the effectiveness of traditional bioinformatics tools in rapidly evolving genomic environments.

Another major challenge is the dependency on large, well-labeled datasets. The process of annotating genomic sequences is both time-consuming and resource-intensive, making it difficult to obtain sufficient labeled data for training robust machine learning models [9], [18]. This issue becomes more critical in scenarios involving rare mutations or newly identified variants, such as those observed in SARS-CoV-2 and other rapidly evolving viruses [5], [7]. The scarcity of labeled data can lead to poor generalization and reduced model performance.

In addition, genomic sequences exhibit complex patterns and long-range dependencies, where interactions between distant nucleotide positions influence biological functions. Traditional machine learning models often rely on handcrafted features and may not effectively capture these dependencies [6], [10]. This results in limited performance when dealing with high-dimensional and sequential genomic data.

Another limitation of existing approaches is their inability to efficiently scale with the increasing size of genomic datasets. As sequencing technologies continue to advance, the volume of genomic data grows exponentially, requiring computational methods that are both scalable and efficient [1], [9]. Conventional methods often become computationally expensive and time-consuming when applied to large-scale datasets.

To address these challenges, this research proposes a hybrid deep learning approach combining DNABERT-2 and Long Short-Term Memory (LSTM) networks. DNABERT-2 leverages transfer learning and transformer-based attention mechanisms to extract contextual representations from DNA sequences without requiring extensive manual feature engineering. It is capable of learning meaningful patterns even with limited labeled data [10], [14].

On the other hand, LSTM networks are designed to model sequential data and capture long-range dependencies within genomic sequences. By integrating DNABERT-2 with LSTM, the proposed model effectively captures both contextual and sequential features, improving mutation classification accuracy. This hybrid framework provides a scalable, efficient, and robust solution for genomic mutation classification, particularly in scenarios involving complex sequence variations and limited labeled data.

1.3 Objectives of the Study

The primary objective of this project is to develop an efficient and scalable mutation classification framework using deep learning techniques, specifically by integrating DNABERT-2 and Long Short-Term Memory (LSTM) networks [1], [6]. The proposed system aims to accurately identify different types of genomic mutations, including point mutations, insertions, deletions, and structural variations, within DNA sequences.

Another key objective is to leverage transfer learning through DNABERT-2 to capture both local and global contextual relationships in genomic sequences [10], [14], [18]. This approach reduces the dependency on large labeled datasets and enhances classification accuracy, particularly in scenarios where annotated data is limited. By utilizing pretrained transformer-based models, the framework aims to improve feature extraction without requiring extensive manual intervention.

The study also aims to incorporate an LSTM-based sequential modeling layer to effectively capture long-range dependencies in DNA sequences. This is essential for detecting mutations that influence distant regions of the genome and for understanding complex

biological interactions within sequences [2].

In addition, the project focuses on developing a model that can generalize across diverse viral genomic datasets, including those obtained from the NCBI Viruses repository. This ensures that the proposed system is capable of handling variations across multiple viral species and genomic structures, making it suitable for real-world applications.

Another objective is to improve the overall performance metrics, such as accuracy and F1-score, by combining contextual and sequential learning approaches. The model is designed to achieve high precision in mutation detection while minimizing false positives and false negatives.

Furthermore, the study aims to create a scalable and computationally efficient framework that can process large genomic datasets [1], [9]. This is important for handling the increasing volume of sequencing data generated in modern genomic research.

Finally, the project seeks to contribute to the advancement of AI-driven bioinformatics by demonstrating the effectiveness of NLP-based deep learning models in genomic mutation classification. The proposed framework is intended to support applications in clinical diagnostics, viral mutation tracking, and large-scale genomic analysis [18], [20].

Chapter 2

Literature Survey

2. LITERATURE SURVEY

2.1 Literature Survey

Recent advancements in deep learning and Natural Language Processing (NLP) have significantly transformed genomic sequence analysis and mutation classification. With the exponential growth of genomic data, traditional computational methods have become insufficient, leading researchers to explore advanced machine learning and deep learning techniques for efficient and scalable analysis.

Transformer-based models have gained widespread attention due to their ability to capture contextual relationships within biological sequences. One of the most prominent models in this domain is DNABERT, which adapts the Bidirectional Encoder Representations from Transformers (BERT) architecture for genomic applications. DNABERT treats DNA sequences as a language by applying k-mer tokenization, allowing the model to learn meaningful representations of nucleotide sequences. Studies have demonstrated that DNABERT achieves superior performance compared to conventional machine learning approaches in tasks such as promoter prediction, transcription factor binding site identification, and mutation classification.

To further improve performance and scalability, DNABERT-2 has been introduced as an enhanced version of DNABERT. This model is designed to efficiently process large-scale genomic datasets and support multi-species genomic analysis. DNABERT-2 incorporates optimized transformer architectures and improved training strategies, enabling better generalization across different genomic structures. As a result, it has shown improved accuracy and robustness in genomic classification tasks.

In parallel, recurrent neural networks (RNNs), particularly Long Short-Term Memory (LSTM) networks, have been widely applied in genomic sequence analysis. LSTM networks are specifically designed to handle sequential data and are capable of capturing long-range dependencies within DNA sequences. This capability is crucial because biological functions often depend on interactions between distant nucleotide positions.

Research has shown that LSTM-based models can effectively learn mutation patterns and improve classification accuracy, especially in sequence-based tasks. Hybrid deep learning approaches have emerged as a promising direction for improving genomic analysis. Researchers have combined transformer-based models with other neural network architectures to leverage their complementary strengths. For example, some studies integrate DNABERT with Convolutional Neural Networks (CNNs) to capture local sequence features and motifs, while others combine DNABERT with LSTM networks to incorporate both contextual and sequential learning. These hybrid models have demonstrated improved performance in various bioinformatics tasks, including enhancer prediction, gene classification, and mutation detection.

Additionally, the application of transfer learning in genomics has gained significant importance. Pretrained models like DNABERT can be fine-tuned for specific tasks, reducing the need for large labeled datasets and improving computational efficiency. This approach is particularly beneficial in genomic research, where annotated data is often limited and expensive to obtain. Transfer learning enables models to leverage knowledge from large pretraining datasets and adapt it to specific mutation classification problems.

Recent studies have also explored the use of genomic foundation models and large language models (LLMs) for biological sequence analysis. These models aim to learn universal representations of genomic data and can be applied to a wide range of tasks, including mutation detection, gene prediction, and disease classification. The use of such models highlights the growing convergence between NLP and bioinformatics.

Despite these advancements, several challenges remain in genomic mutation classification. Many existing models struggle with class imbalance, where certain mutation types are underrepresented, leading to biased predictions. Limited availability of labeled data further complicates model training and evaluation. Additionally, some models focus primarily on contextual learning while failing to capture long-range sequential dependencies, which are critical for accurate genomic analysis.

To address these limitations, the present study proposes a hybrid DNABERT-2 and LSTM model that combines transformer-based contextual learning with sequential modeling. This approach aims to improve mutation classification accuracy, enhance generalization across diverse genomic datasets, and provide a scalable and efficient solution for analyzing complex genomic sequences.

The following table summarizes the key approaches reviewed:

Table 2.1: Summary of Literature Survey Approaches

Year	Author	Model	Key Focus	Gap
2023	Ma et al.	Transformer + Mamba2	Long genomic sequences	Increased model complexity
2023	Zhou et al.	DNABERT-S	Metagenomic classification	No mutation-level analysis
2022	Zhang et al.	DNABERT	Viral genome classification	No evolution modeling
2022	Lee et al.	DNABERT + Contrastive Learning	Representation learning	No temporal dependency
2022	Kumar et al.	DNABERT	Disease mutation analysis	Not tested on viral data
2021	Wu et al.	DNABERT	Cancer mutation detection	Dataset-specific results
2021	Ji et al.	DNABERT + CNN	Promoter prediction	Limited long-range dependency capture
2020	Devlin et al.	BERT	Language representation learning	Not designed for genomic data

Table represents a comparative analysis of existing research works related to genomic sequence analysis and mutation classification, highlighting the models used, their key focus areas, and associated limitations. Earlier studies primarily focused on transformer-based models such as BERT and DNABERT for learning contextual representations of DNA sequences. While these models are effective in capturing local and contextual relationships, they often lack the ability to model long-range dependencies. Some approaches combined DNABERT with techniques like CNN or contrastive learning, but these methods still faced challenges in capturing temporal dependencies and handling complex genomic patterns.

Recent advancements, including DNABERT-S and transformer-based architectures, have improved performance in tasks such as metagenomic classification and long-sequence analysis. However, issues such as increased computational complexity, lack of mutation-level analysis, and limited generalization across diverse datasets remain. Many existing models are also dataset-specific and do not effectively handle rapidly evolving viral genomes. To address these limitations, the proposed work utilizes a hybrid DNABERT-2 and LSTM model, which combines contextual feature extraction with sequential modeling to improve mutation classification accuracy and provide better generalization across multiple viral datasets.

Chapter 3

System Analysis & Feasibility

Study

3. SYSTEM ANALYSIS & FEASIBILITY STUDY

3.1 Existing System

This is based on the iEnhancer-BERT model, a transfer learning framework developed for enhancer identification and classification in genomic sequences [14], [18]. This model combines a DNABERT-based transformer architecture with a Convolutional Neural Network (CNN) to improve feature extraction and classification performance. The system begins with preprocessing DNA sequences using k-mer tokenization, where each sequence is divided into overlapping subsequences (k-mers) that act as input tokens for the model.

These tokenized sequences are passed through a pre-trained DNABERT model, which is trained on the human genome and is capable of capturing rich contextual relationships between nucleotides using self-attention mechanisms [10], [18]. Unlike traditional approaches that rely only on the final output of the transformer, iEnhancer-BERT utilizes outputs from all transformer encoder layers to construct a comprehensive feature representation. This multi-layer feature extraction improves the model's ability to learn complex genomic patterns.

The extracted features are then fed into a CNN layer, which performs further feature extraction through convolution and pooling operations. The CNN captures local sequence patterns and motifs, while the final classification is carried out using fully connected layers and a softmax function. The model is trained using optimization techniques such as binary cross-entropy loss, ensuring accurate classification results.

The system follows a two-stage identification framework. In the first stage, the model classifies DNA sequences as enhancers or non-enhancers. In the second stage, the identified enhancers are further classified into strong or weak enhancers, enabling more detailed analysis of genomic regulatory elements. This structured approach improves classification accuracy by dividing the task into manageable steps [17].

Despite achieving strong performance compared to earlier models such as iEnhancer-2L and BERT-2D, the existing system has several limitations. It primarily focuses on enhancer identification rather than mutation classification, limiting its applicability in broader genomic analysis. The model also relies on fixed-length sequences (e.g., 200 base pairs), which may not represent real-world genomic data that varies in length. Furthermore, while CNN effectively captures local features, it is less effective in modeling long-range dependencies across DNA sequences [6], [10].

Another limitation is the lack of sequence evolution modeling and cross-species generalization, which are important for analyzing viral genomes and mutation patterns [4], [22]. These challenges indicate the need for more advanced hybrid models that can integrate contextual learning with sequential dependency modeling, motivating the development of improved architectures such as DNABERT-2 combined with LSTM networks.

3.1.1 Limitations of Existing System

The existing system, iEnhancer-BERT, is primarily designed for enhancer identification and classification rather than genomic mutation analysis. This limits its applicability in broader genomic studies, especially in tasks such as mutation detection and viral genome analysis [14], [18]. Additionally, the model processes DNA sequences of fixed length (e.g., 200 base pairs), whereas real-world genomic data often consists of variable-length sequences. This restriction reduces the flexibility of the system when applied to diverse biological datasets.

Another major limitation is its inability to effectively capture long-range dependencies in DNA sequences. Although DNABERT provides contextual representations, the CNN layer mainly focuses on local feature extraction and does not fully model sequential relationships across distant regions of the genome [6], [10]. Furthermore, the combination of transformer-based models and CNN layers results in high computational complexity, making the system resource-intensive and less efficient for large-scale genomic analysis [1], [9].

The model also depends heavily on pretrained DNABERT, which is trained on specific genomic datasets, potentially affecting its performance when applied to new or diverse data sources. In addition, the system shows limited generalization across different species, as it is primarily trained on human genomic data. It also lacks the ability to explicitly model evolutionary patterns and mutation dynamics, which are essential for understanding viral mutations and disease progression [5], [22]. Moreover, the two-stage classification framework increases system complexity and may lead to error propagation between stages, affecting overall performance [17].

3.2 Proposed System

The proposed system introduces a hybrid deep learning framework that combines DNABERT-2, a transformer-based DNA language model, with a Long Short-Term Memory (LSTM) network for accurate mutation classification in viral genomic sequences. Unlike traditional alignment-based approaches, this system treats DNA sequences as a form of biological language and applies Natural Language Processing (NLP) techniques to capture complex genomic patterns [6], [20]. Initially, viral genomic sequences are collected from the NCBI Viruses database and labeled as either Wildtype or Mutated to perform binary classification. These sequences are then transformed into overlapping k-mers, which convert raw nucleotide sequences into tokenized representations suitable for deep learning models [18].

The tokenized sequences are processed using the DNABERT-2 model, which generates high-dimensional contextual embeddings through bidirectional self-attention mechanisms. This allows the model to capture both local sequence motifs and global contextual relationships within DNA sequences [10], [14]. By leveraging transfer learning, DNABERT-2 utilizes knowledge from large-scale genomic pretraining, enabling improved performance even when labeled data is limited. The embeddings generated by DNABERT-2 serve as rich feature representations for further analysis.

To enhance sequential learning, the extracted embeddings are passed into an LSTM network, which is specifically designed to model long-range dependencies in sequential data. The LSTM layer captures temporal relationships and maintains contextual memory across the sequence, enabling the system to detect mutation patterns that may span distant regions of the genome [2], [10]. This combination of DNABERT-2 and LSTM ensures that

both contextual and sequential features are effectively learned.

Finally, the output from the LSTM layer is fed into a fully connected classification layer, where a sigmoid activation function is applied to generate a probability score. Based on this score, the sequence is classified as either wildtype (0) or mutated (1). The model is trained and fine-tuned using labeled viral datasets, with optimization techniques such as binary cross-entropy loss, weighted loss functions, and early stopping to improve generalization and prevent overfitting.

Overall, the proposed DNABERT-2 + LSTM hybrid architecture provides a scalable, accurate, and robust framework for genomic mutation classification. By integrating contextual learning with sequential modeling, the system overcomes the limitations of existing methods and offers an effective solution for analyzing complex viral genomic data.

3.2.1 Advantages of Proposed System:

The proposed DNABERT-2 and LSTM hybrid model offers several advantages over traditional and existing genomic analysis methods. One of the key benefits is its ability to capture both contextual and sequential features of DNA sequences. While DNABERT-2 effectively learns contextual relationships using transformer-based attention mechanisms, the LSTM network models long-range dependencies, enabling accurate detection of mutation patterns across distant regions of the genome [10], [14].

Another major advantage is the use of transfer learning, where the pretrained DNABERT-2 model leverages knowledge from large-scale genomic data. This reduces the dependency on large labeled datasets and improves performance even with limited training data [9], [18]. As a result, the system becomes more efficient and adaptable to different genomic datasets, including viral sequences.

The proposed system is also highly scalable and efficient for handling large genomic datasets. Unlike traditional alignment-based methods, it does not rely on reference genomes, making it suitable for analyzing new and evolving viral strains. This makes the model particularly useful in real-world applications such as mutation tracking, disease analysis, and outbreak monitoring.

In addition, the hybrid architecture improves classification accuracy and robustness by combining the strengths of transformer models and recurrent neural networks. The model minimizes errors such as false positives and false negatives, leading to better performance in mutation classification tasks. The use of optimization techniques further enhances generalization and prevents overfitting.

Finally, the system provides a more flexible and biologically meaningful approach to genomic analysis. By treating DNA sequences as a language and applying NLP techniques, the model can uncover complex patterns that are difficult to detect using conventional methods [6], [20]. This makes the proposed framework a powerful tool for advanced genomic research and bioinformatics applications.

3.3 Dataset

The nucleotide datasets from the NCBI Viruses database compile extensive viral sequence data from GenBank and other biological repositories. These datasets provide a rich source of genomic information for computational analysis and are widely used in bioinformatics research. They are particularly useful for applying Natural Language Processing (NLP) techniques in genomics, where DNA sequences are treated as a form of biological language. Using models such as DNABERT and LSTM, these datasets support effective mutation classification through transfer learning. This enables the extraction of meaningful patterns, improves prediction accuracy, and facilitates large-scale genomic analysis.

Human Viruses Dataset (22,894 entries): The dataset contains nucleotide sequences of human infecting viruses. In genomic studies, this data can help scientists study viral mutations, evolutionary trends, and host/virus interaction. The researchers can apply the DNA-BERT model, which is basically a transformer model, to the genomic data as a valuable genomic insight tool for mutation classification and disease prediction. At the same time, the LSTM (Long Short-Term Memory) networks enable sequence processing by modeling long-term dependencies within the viral genome. This dataset is helpful in mutation tracking, which plays an essential role in vaccine development and developing antiviral drugs.

Influenza Viruses Dataset (40,805 entries): This dataset represents nucleotide sequences among different strains of Influenza A and B viruses. Influenza viruses are responsible for seasonal outbreaks and pandemics due to their ability to mutate rapidly. DNA-BERT models in NLP sequencing enhance the classification of mutations based on recognizing, while the temporal dependencies examined by LSTM networks improve predictions in the viral evolution.

The dataset is crucial in monitoring influenza and allowing researchers to predict new strains with which they can hopefully develop vaccines. Genetic analysis has been significantly advanced through transfer learning-based models, providing a deeper comprehension of antigenic drift and shift.

SARS-CoV-2 Dataset of Viruses (3,604 entries): The SARS-CoV-2 dataset lies within the nucleotide sequences of the virus that caused COVID-19. Given that this virus has a specific rate of mutation so much so that one would find it in an entirely new variant by the time he or she gets to the final stages of his or her studies, this dataset is of great value in tracking variants and evaluating their transmissibility and effects on immune evasion.

In mutation analysis within advanced deep learning upon a genome sequence analysis, DNA-BERT does the work, while LSTM networks capture sequence dependency in the viral evolution. This library has been instrumental in variant detection and informing public health and vaccination strategies. The application of language transfer learning models in genomics has totally changed SARS-CoV-2 research by making real-time mutation tracking and predictive modeling a practical reality.

The integration of these datasets hence pushes genomics research forward, providing foundations for mutation classification, vaccine development, and epidemiological studies. The combination of DNA BERT with LSTM should unravel viral evolution in order to formulate better strategies for preventing and treating diseases.

Table 3.1: Dataset Description

Dataset Name	Source	No. of Entries	Description	Application
Human Viruses Dataset	NCBI Viruses (GenBank)	22,894	Contains nucleotide sequences of viruses that infect humans. Useful for studying mutation patterns, viral evolution, and host-virus interactions.	Mutation classification, disease prediction, vaccine and antiviral drug development.
Influenza Viruses Dataset	NCBI Viruses (GenBank)	40,805	Includes sequences of Influenza A and B virus strains. Known for rapid mutation causing seasonal outbreaks and pandemics.	Strain prediction, mutation tracking, antigenic drift and shift analysis, vaccine development.
SARS-CoV-2 Dataset	NCBI Viruses (GenBank)	3,604	Includes sequences of Influenza A and B virus strains. Known for rapid mutation causing seasonal outbreaks and pandemics.	Variant detection, mutation analysis, public health monitoring, vaccine strategy.

3.4 Data Pre-Processing

Data preprocessing is an essential step to transform raw genomic sequences into a format suitable for deep learning models. After collecting sequences from the NCBI Viruses database, the data is first standardized to maintain consistency in format across all samples. This ensures uniform input for the model.

Next, long DNA sequences are divided into fixed-length segments to handle variable-length data and improve computational efficiency. The segmented sequences are then converted into k-mer representations, where each sequence is broken into overlapping subsequences of length k . This step allows the model to treat DNA sequences like words in natural language.

Finally, the k-mer sequences are tokenized and encoded into numerical vectors using a compatible tokenizer. These encoded sequences are then provided as input to the DNABERT-2 model and further processed by the LSTM layer for mutation classification.

3.4.1 Data Cleaning

Data cleaning is performed to ensure the quality and reliability of the dataset before preprocessing. The raw genomic data may contain duplicate, incomplete, or noisy sequences, which can negatively impact model performance.

Initially, duplicate sequences are removed to avoid bias during training. Then, incomplete or corrupted sequences are filtered out. The dataset is also checked for invalid or ambiguous nucleotide characters (such as N, R, Y), and only valid nucleotides (A, T, G, C) are retained.

Additionally, any unnecessary information or irrelevant data is removed, ensuring that only clean and meaningful genomic sequences are used. This step improves data consistency and enhances the overall effectiveness of the proposed model.

3.4.2 Feature Understanding and Transformation

Feature understanding and transformation play a crucial role in converting raw genomic sequences into meaningful representations for deep learning models. In genomic data, the primary features are nucleotide sequences (A, T, G, C), which carry biological information related to gene structure, mutations, and functional elements. Understanding these sequences involves identifying patterns, motifs, and dependencies that influence mutation behavior [6], [20].

To make these features suitable for machine learning, the raw DNA sequences are transformed using k-mer representation, where sequences are divided into overlapping substrings of length k. This transformation helps in capturing local sequence patterns and provides a structured way to represent genomic data, similar to words in natural language processing [18].

After k-mer generation, the sequences are passed through a tokenization process, where each k-mer is converted into a token and mapped to a numerical index. These tokens are then transformed into dense vector embeddings using the DNABERT-2 model. The embeddings capture both contextual relationships and semantic meaning of nucleotide sequences through attention mechanisms [10], [19].

Finally, these transformed features are fed into the LSTM network, which further processes the embeddings to capture sequential dependencies and long-range relationships within the DNA sequence. This combined feature transformation process enables the model to effectively learn complex genomic patterns and improve mutation classification accuracy.

3.4.3 Feature Scaling

Feature scaling is an important step in data preprocessing to ensure that input features are on a similar scale, which helps improve the performance and stability of machine learning models. In traditional machine learning, techniques such as normalization and standardization are commonly used to scale numerical features.

In genomic sequence analysis, raw DNA data consists of categorical values (A, T, G, C) rather than numerical features. Therefore, direct feature scaling is not applied to the raw sequences. Instead, after k-mer tokenization and encoding, the sequences are converted into numerical representations and then into dense embeddings using the DNABERT-2 model. These embeddings are already normalized and structured in a way that makes them suitable for deep learning models.

Additionally, during model training, techniques such as layer normalization and batch normalization are used within the neural network to maintain stable feature distributions. These methods help in faster convergence, reduce internal covariate shift, and improve overall model performance.

Thus, while traditional feature scaling methods are not directly applied to genomic sequences, the use of embedding representations and normalization techniques ensures that the features are appropriately scaled for effective learning.

3.4.4 Handling Class Imbalance

Class imbalance is a common issue in genomic datasets, where the number of samples in one class (e.g., mutated sequences) may be significantly lower than the other class (wildtype sequences). This imbalance can lead to biased model predictions, where the model tends to favor the majority class and performs poorly on the minority class [9], [18].

To address this issue, several techniques are applied in the proposed system. One effective approach is the use of class weighting, where higher importance is assigned to the minority class during training. This ensures that misclassification of minority class samples is penalized more, encouraging the model to learn balanced decision boundaries [1], [9].

Another method involves data resampling techniques, such as oversampling the minority class or undersampling the majority class. Oversampling methods, including techniques like duplication or synthetic data generation, help increase the representation of the minority class, while undersampling reduces the dominance of the majority class [18].

Additionally, the model uses evaluation metrics such as precision, recall, and F1-score instead of relying only on accuracy. These metrics provide a better understanding of model performance in imbalanced datasets, especially in correctly identifying minority class samples.

Furthermore, techniques like early stopping and regularization are applied to prevent overfitting toward the majority class. By combining these strategies, the proposed system effectively handles class imbalance and improves the reliability and fairness of mutation classification.

3.4.5 Train-Test Split

The dataset is divided into training and testing sets to evaluate the performance of the proposed model. This step ensures that the model is trained on one portion of the data and evaluated on unseen data, helping to measure its generalization ability.

Typically, the dataset is split in a ratio such as 80:20, where 80% of the data is used for training and 20% is used for testing. The training set is used to learn patterns and optimize model parameters, while the testing set is used to assess the model's performance on new data.

To maintain consistency and avoid bias, the split is performed using a random sampling method, ensuring that both training and testing sets contain a balanced distribution of classes (wildtype and mutated). In some cases, stratified sampling is applied to preserve the proportion of each class in both sets, especially when dealing with imbalanced datasets.

Additionally, a portion of the training data may be further divided into a validation set for tuning hyper-parameters and preventing overfitting. This helps in selecting the best model configuration. Overall, the train-test split ensures reliable model evaluation and helps in building a robust mutation classification system.

3.5 Libraries Utilized

Pandas is a powerful Python library used for data manipulation and analysis. It is widely applied in handling large genomic datasets, especially nucleotide sequence data. In genomic NLP tasks, Pandas helps convert raw DNA sequences into structured DataFrames, enabling efficient filtering, cleaning, and transformation of data. It supports batch processing and simplifies handling large-scale datasets, making it easier to prepare genomic sequences for mutation classification using models like DNABERT and LSTM.

NumPy is a fundamental library for numerical computing in Python and plays a key role in genomic data processing. It provides efficient support for multi-dimensional arrays and matrix operations, which are essential for converting nucleotide sequences into numerical formats. NumPy enables fast computation and is widely used for encoding genomic data into embeddings suitable for deep learning models. Its compatibility with frameworks like PyTorch enhances performance in large-scale genomic analysis.

The Transformers library (Hugging Face) is central to implementing language models in genomics. It provides pretrained models such as DNABERT, which are specifically designed to process DNA sequences using self-attention mechanisms. This allows the model to capture complex relationships within genomic data. The library also supports fine-tuning on specific datasets and generates contextual embeddings, which are further used by sequential models like LSTM for mutation classification tasks.

PyTorch is used for building, training, and optimizing deep learning models. It supports efficient tensor operations and GPU acceleration, making it suitable for handling large genomic datasets. The `torch.utils.data` module provides tools for data loading, batching, and shuffling, which are essential for training models. PyTorch also offers built-in layers such as Embedding, LSTM, and Linear layers, enabling the development of hybrid architectures like DNABERT-2 with LSTM. Additionally, optimization algorithms like Adam and SGD are used to minimize loss and improve model performance during training.

3.6 Methodology

This study proposes a hybrid deep learning framework that combines the contextual learning capability of transformer-based models with the sequential modeling strength of recurrent neural networks for genomic mutation classification. The proposed system integrates DNABERT-2 for feature extraction and Long Short-Term Memory (LSTM) networks for sequential learning. This hybrid approach enables the model to effectively capture both local contextual relationships and long-range dependencies present in DNA sequences, which are essential for accurate mutation detection [10], [14].

Unlike traditional approaches, the proposed framework treats DNA sequences as a form of biological language and applies Natural Language Processing (NLP) techniques to analyze genomic patterns [6]. DNABERT-2 leverages bidirectional self-attention mechanisms to learn meaningful representations of nucleotide sequences, while the LSTM layer enhances the model’s ability to preserve sequential information and temporal dependencies across long genomic regions. This combination improves the model’s capability to detect complex mutation patterns and enhances overall classification performance.

In addition, the use of transfer learning allows DNABERT-2 to utilize knowledge from large-scale genomic datasets, reducing the need for extensive labeled data and improving learning efficiency [9], [18]. The hybrid architecture also enhances model generalization, robustness, and adaptability when applied to diverse viral genomic datasets. Furthermore, the integration of transformer-based contextual learning with sequential modeling helps in minimizing information loss and capturing both global and temporal features effectively.

The overall workflow of the system consists of multiple stages, including dataset collection, data preprocessing, k-mer tokenization, feature extraction using DNABERT-2, sequential modeling using LSTM, and final classification. During preprocessing, raw DNA sequences are cleaned, normalized, and converted into structured formats. The k-mer tokenization step transforms sequences into overlapping subsequences, enabling compatibility with transformer models [12].

Following this, DNABERT-2 generates high-dimensional contextual embeddings, which serve as input to the LSTM layer. The LSTM processes these embeddings sequentially to

capture long-term dependencies and positional relationships within the genomic data. The final output from the LSTM is passed through a fully connected dense layer, where a sigmoid activation function is used to perform binary classification (wildtype or mutated).

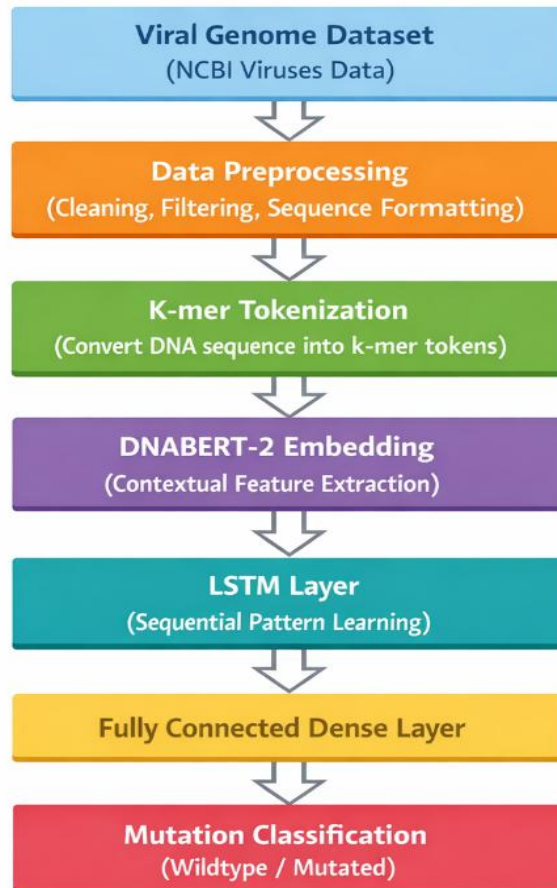


Figure 3.1: Methodology Overview

To improve model performance, techniques such as batch training, learning rate optimization, dropout regularization, and early stopping are applied during training. Evaluation metrics including accuracy, precision, recall, and F1-score are used to assess the effectiveness of the model. This comprehensive methodology ensures that the proposed system is scalable, efficient, and capable of handling large-scale genomic datasets, making it suitable for real-world applications such as mutation detection, viral genome analysis, and disease prediction [5].

3.6.1 Auto-Tokenizer

For mutation classification in genomics using NLP techniques, models such as DNABERT and LSTM utilize the AutoTokenizer from the Hugging Face Transformers library to preprocess DNA sequences. Since DNA sequences consist of nucleotide bases (A, T, G, C), they require specialized tokenization rather than traditional natural language processing. DNABERT applies k-mer tokenization, which divides sequences into overlapping substrings of fixed length, enabling efficient processing of large genomic datasets such as those from NCBI without manual intervention [18]. This approach ensures a consistent, structured, and automated numerical representation suitable for downstream embedding layers.

After tokenization, sequences are converted into input IDs and attention masks for transformer-based processing. Input IDs represent numerical indices corresponding to each k-mer token, while attention masks help distinguish valid tokens from padding tokens when sequences vary in length. This mechanism ensures efficient parallel batch processing and prevents the model from learning from irrelevant padded values, thereby improving both training stability and model accuracy [10], [14].

The AutoTokenizer integrates seamlessly with DNABERT, allowing easy fine-tuning on domain-specific genomic datasets. It supports advanced features such as dynamic padding, truncation strategies, and maximum sequence length control, which help optimize memory usage and computational efficiency. Special tokens such as [CLS] (classification token) and [SEP] (separator token) are automatically added to define sequence boundaries and provide additional contextual signals for the transformer model [10].

In addition, AutoTokenizer ensures vocabulary consistency by mapping k-mers to a predefined token dictionary, which is essential for maintaining compatibility with pretrained DNABERT models. It also supports batch encoding, enabling the processing of multiple sequences simultaneously, which significantly speeds up preprocessing for large-scale genomic datasets [5], [16]. This is particularly important when working with high-throughput sequencing data.

Furthermore, the tokenizer pipeline allows for efficient handling of unknown or rare k-mers, ensuring robustness when dealing with unseen genomic variations. The embeddings generated after tokenization are passed to the DNABERT model, which produces contextual representations that are further processed by the LSTM layer to capture long-range dependencies and sequential mutation patterns [2], [10].

Overall, the use of AutoTokenizer provides an automated, scalable, and efficient preprocessing framework, reducing manual effort while ensuring high-quality input representation. It plays a critical role in bridging raw genomic data with deep learning models, ultimately enhancing the performance of mutation classification systems.

3.6.2 Positional Encoding

Positional encoding is a fundamental component of transformer architectures, enabling the model to incorporate sequence order information into input embeddings. Since transformers do not inherently process data sequentially like recurrent models, positional encoding provides a mechanism to explicitly represent the position of each token within a sequence. It is typically represented as a matrix that is added to the input embedding vectors, ensuring that positional information is preserved throughout the network [7].

The positional encoding matrix (PE) is generated for a sequence of length L and embedding dimension d , resulting in a matrix of size $L \times d$. Each row in this matrix corresponds to a positional encoding vector for a specific token in the sequence. These vectors are computed using sine and cosine functions of varying frequencies, allowing the model to encode both absolute positions (exact location) and relative positions (distance between tokens). This design ensures that each position has a unique representation while maintaining smooth transitions between neighboring positions [6], [15].

In this formulation, pos represents the position of the token in the sequence, i denotes the dimension index, and d refers to the embedding size. The use of alternating sine and cosine functions ensures that positional encodings are continuous, non-learned, and generalizable, meaning the model can handle sequences longer than those encountered during training.

Additionally, these encodings allow the transformer to easily learn positional relationships through linear transformations, improving efficiency.

The positional encoding matrix is combined with the input embeddings using element-wise addition, integrating positional and semantic information into a unified representation. This enriched embedding is then processed by the self-attention mechanism, where the model can simultaneously consider both the content of tokens and their positions. This is crucial for accurately modeling dependencies in sequences [17], [23].

Furthermore, positional encoding improves the model’s ability to capture order-sensitive patterns, which is particularly important in genomic data. In DNA sequences, the position of nucleotides directly influences biological functions such as gene expression, transcription factor binding, and mutation impact. By encoding positional information, models like DNABERT can effectively identify sequence motifs, regulatory regions, and mutation hotspots.

$$PE_{pos,2i} = \sin\left(\frac{pos}{10000^{\frac{2i}{d}}}\right)$$

$$PE_{pos,2i+1} = \cos\left(\frac{pos}{10000^{\frac{2i}{d}}}\right)$$

Another important aspect is that positional encoding helps reduce ambiguity in attention mechanisms by ensuring that tokens with similar embeddings but different positions are treated distinctly. It also supports parallel processing, which is a key advantage of transformer models over traditional sequential architectures like RNNs and LSTMs.

In the context of genomic sequence analysis, positional encoding plays a critical role in helping models understand the structural organization of DNA sequences. It enables the detection of long-range interactions between distant nucleotide positions, which are essential for mutation classification and evolutionary analysis.

Overall, positional encoding is essential for enabling transformers to process sequential data effectively. By embedding positional information into token representations, it bridges

the gap between self-attention mechanisms and sequence order, making it a key component in both natural language processing and genomic sequence analysis tasks [11], [13].

3.7 BERT Architecture

BERT (Bidirectional Encoder Representations from Transformers) is a powerful deep learning model developed by Google that has significantly transformed the field of Natural Language Processing (NLP). Its key innovation lies in its ability to understand bidirectional context, meaning it analyzes text both from left-to-right and right-to-left simultaneously. Unlike traditional models that process text in a single direction, BERT captures the full context of a word by considering all surrounding words in a sentence. This capability enables it to model complex relationships and semantic meanings more effectively, leading to major improvements in tasks such as text classification, question answering, and named entity recognition [19], [21].

BERT is built using a stack of transformer encoder layers, where each layer consists of multi-head self-attention mechanisms and position-wise feed-forward neural networks. The self-attention mechanism allows the model to focus on different parts of the input sequence simultaneously, learning contextual relationships between tokens regardless of their distance [14]. Multi-head attention further enhances this capability by enabling the model to attend to multiple representation subspaces in parallel. Each transformer layer also includes residual connections and layer normalization (Add & Norm), which help stabilize training, reduce vanishing gradients, and improve convergence.

The architecture also incorporates input embeddings, segment embeddings, and positional encodings. Input embeddings represent the tokens, segment embeddings distinguish between sentence pairs, and positional encodings provide information about token order. These embeddings are combined and passed through the transformer layers, enabling BERT to process both semantic and positional information effectively [15], [17].

A key strength of BERT is its pre-training strategy, which involves training on large-scale text corpora using two main objectives: Masked Language Modeling (MLM) and Next Sentence Prediction (NSP). In MLM, a percentage of input tokens are randomly masked,

and the model learns to predict them using contextual information. NSP trains the model to determine whether one sentence logically follows another. These objectives allow BERT to learn deep contextual representations and sentence-level relationships.

Another important aspect of BERT is the use of a special [CLS] token, which is added at the beginning of every input sequence and serves as a global representation for classification tasks. Similarly, the [SEP] token is used to separate sentences. During fine-tuning, the output corresponding to the [CLS] token is typically fed into a classification layer to perform downstream tasks [12], [21].

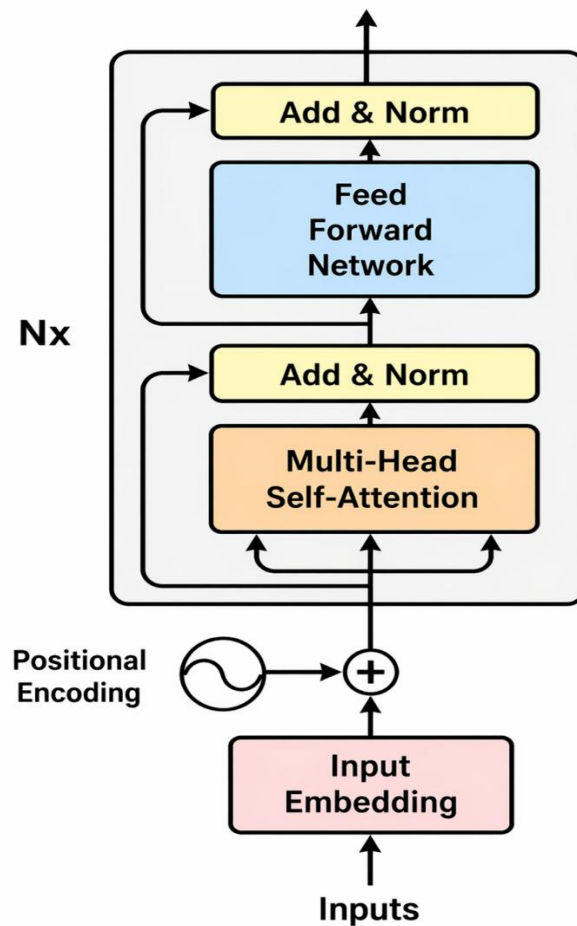


Figure 3.2: BERT Architecture

One of the major advantages of BERT is its ability to be fine-tuned for specific tasks with minimal additional training. This makes it highly adaptable across domains. In genomics,

models such as DNABERT extend the BERT architecture to analyze DNA sequences by treating nucleotide sequences (A, T, G, C) as a form of language [4], [8]. This enables the model to identify patterns, motifs, and mutations within genomic data.

To further enhance sequence modeling, BERT-based embeddings are often combined with Long Short-Term Memory (LSTM) networks, which are effective in capturing long-range dependencies in sequential data. While BERT captures contextual relationships, LSTM preserves sequential order and temporal dependencies, making the hybrid approach highly effective for mutation classification and genomic analysis [14], [15].

Moreover, BERT has expanded beyond traditional NLP into domains such as healthcare, biomedical research, and bioinformatics, enabling advancements in disease prediction, drug discovery, and genetic analysis. Its ability to model complex contextual relationships, process large-scale data, and adapt to different applications has made it a cornerstone of modern AI systems. As a result, BERT and its variants continue to play a crucial role in advancing language-based deep learning applications, including genomic sequence analysis and mutation detection.

3.8 Impact of DNABERT

DNABERT is a specialized adaptation of the BERT architecture designed for genomic data, particularly DNA sequences. While traditional BERT was developed for natural language processing, DNABERT applies the same bidirectional transformer principles to DNA by treating fixed-length substrings called k-mers as tokens. Since DNA consists of four nucleotides (A, T, C, G), k-mer tokenization converts sequences into overlapping substrings, enabling the model to process them similarly to language data and capture meaningful biological patterns [18], [14].

DNABERT is pre-trained using masked language modeling (MLM), where random k-mers are masked and predicted based on surrounding context. This training strategy allows the model to learn rich contextual representations of DNA sequences, capturing both short-range motifs and long-range dependencies that are essential for biological functions such as gene regulation, mutation impact analysis, and protein-DNA interactions. The

bidirectional self-attention mechanism enables the model to understand relationships between upstream and downstream regions of DNA, significantly improving performance in tasks like mutation classification and promoter prediction [10], [20].

Built on the BERT-base architecture, DNABERT typically consists of 12 transformer layers with multi-head self-attention and a hidden size of 768 dimensions. During fine-tuning, a task-specific classification layer is added, allowing the model to adapt to specific genomic tasks while retaining knowledge learned during pretraining. This transfer learning capability reduces the need for large labeled datasets, lowers computational cost, and accelerates model deployment in real-world genomic applications [9], [17].

The impact of DNABERT in bioinformatics is substantial, as it eliminates the need for manual feature engineering by automatically learning complex sequence patterns through attention mechanisms. It provides higher accuracy, scalability, and efficiency compared to traditional machine learning and alignment-based approaches. DNABERT has been successfully applied in tasks such as mutation classification, enhancer detection, gene function prediction, regulatory element identification, and transcription factor binding site analysis.

Furthermore, DNABERT has contributed to the emergence of genomic foundation models, enabling large-scale and cross-species analysis of biological data. It supports multi-task learning, where a single pretrained model can be fine-tuned for multiple genomic tasks, improving research efficiency. Its ability to generalize across different datasets makes it highly suitable for studying diverse organisms and viral genomes [5], [7].

DNABERT plays a critical role in viral mutation tracking and evolutionary analysis, particularly for rapidly evolving viruses such as SARS-CoV-2. It enables researchers to identify mutation patterns, predict functional impacts, and monitor genomic variations in near real-time. This has significant implications for public health, vaccine development, and outbreak management [20].

In addition, DNABERT supports advancements in precision medicine, where genomic data

is used to tailor treatments based on individual genetic profiles. It aids in identifying disease-associated mutations, understanding genetic disorders, and improving diagnostic accuracy. The model also contributes to drug discovery and biomarker identification, accelerating biomedical research [23].

By bridging the gap between NLP and genomics, DNABERT has introduced a new paradigm of treating biological sequences as language, enabling deeper insights into genomic structures and functions. Its integration with models like LSTM further enhances sequential learning, allowing the system to capture both contextual and temporal dependencies. This hybrid approach makes DNABERT a powerful and versatile tool for analyzing complex genomic data and significantly improving mutation classification performance.

3.9 Long Short-Term Memory

Recent studies have explored the use of Long Short-Term Memory (LSTM) networks in genomic analysis, particularly in combination with language-based transfer learning models such as DNABERT for mutation prediction. DNABERT generates deep contextual embeddings of DNA sequences using transformer-based self-attention mechanisms, enabling the model to capture relationships between nucleotides within a sequence. However, genomic data often exhibits long-range dependencies, where mutations at distant positions can influence gene expression, regulatory functions, and overall biological outcomes [5], [7]. LSTM, a specialized type of recurrent neural network (RNN), complements DNABERT by effectively modeling temporal, sequential, and order-dependent relationships within DNA sequences.

After DNA sequences are tokenized into k-mers and transformed into contextual embeddings by DNABERT, the LSTM layer processes these embeddings sequentially. This allows the model to capture order-sensitive information and positional dependencies that may not be fully represented by transformer models alone. While DNABERT focuses on contextual and semantic relationships, LSTM ensures that the sequential flow and structural continuity of genomic information are preserved, making the hybrid architecture more effective for mutation classification [9], [14].

The LSTM network is built using memory cells and gating mechanisms, including the input gate, forget gate, and output gate. These gates regulate the flow of information by deciding what information to retain, update, or discard at each time step. This design helps overcome common issues in traditional RNNs such as the vanishing gradient problem, enabling LSTM to maintain information over long sequences. This selective memory mechanism is particularly important in genomic analysis, where only specific regions of DNA contribute to functional or mutational changes.

Another important feature of LSTM is its ability to capture long-distance interactions and dependencies between nucleotides [15]. In biological systems, interactions between distant regions of DNA can significantly impact gene expression and regulatory processes. LSTM effectively models these interactions by maintaining a continuous memory state, allowing the network to detect subtle mutation patterns and hidden relationships across the sequence.

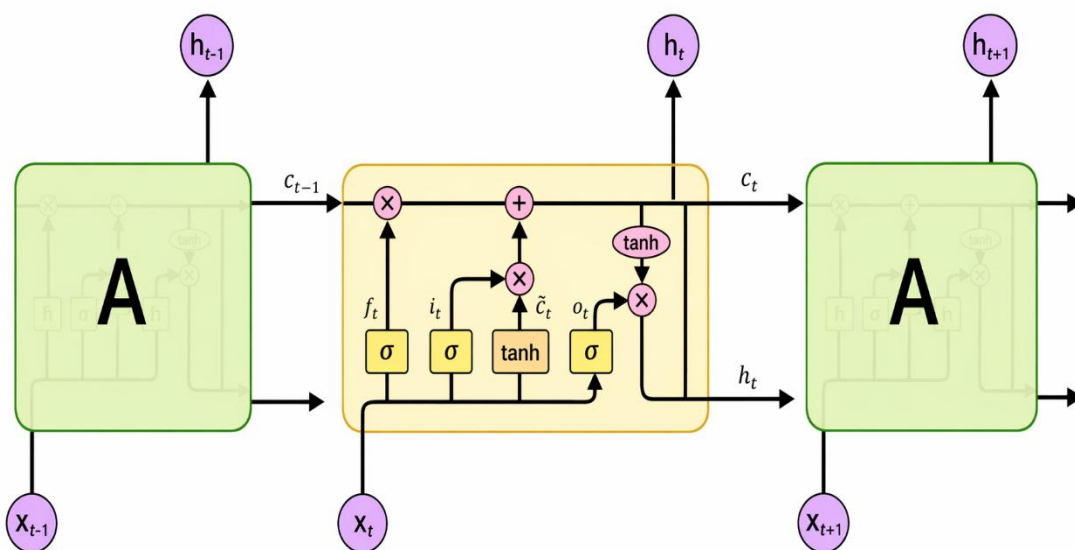


Figure 3.3: LSTM Network

Additionally, LSTM supports variable-length sequence processing, which is essential for real-world genomic datasets such as those from the NCBI Viruses repository. This flexibility improves the model's ability to generalize across different organisms, viral

strains, and sequence structures. LSTM also plays a key role in capturing temporal evolution patterns in viral genomes, helping researchers understand how mutations accumulate and evolve over time [2], [11].

Furthermore, LSTM enhances the robustness, stability, and generalization capability of the hybrid model by refining the high-level embeddings generated by DNABERT. It reduces noise, minimizes information loss, and improves feature representation, leading to better classification performance. The combination of contextual embeddings from DNABERT and sequential learning from LSTM results in a more comprehensive understanding of genomic data.

The integration of DNABERT and LSTM enables efficient learning from large-scale genomic datasets, making the model suitable for applications such as mutation classification, viral genome analysis, evolutionary studies, disease prediction, and precision medicine. It also supports the identification of mutation hotspots, regulatory regions, and functional genomic elements [19].

Overall, the combination of DNABERT and LSTM provides a powerful hybrid framework that leverages both contextual understanding and sequential learning, making it highly effective for analyzing complex genomic data and significantly improving mutation classification performance.

3.10 Model Validation

Recent studies have explored the use of Long Short-Term Memory (LSTM) networks in genomic analysis, especially when combined with models like DNABERT for mutation prediction. DNABERT generates contextual embeddings using self-attention, capturing relationships between nucleotides. However, genomic data often contains long-range dependencies, where distant mutations influence biological functions. LSTM complements DNABERT by modeling sequential and temporal relationships within DNA sequences.

After k-mer tokenization and embedding through DNABERT, the LSTM layer processes the data sequentially to capture order-sensitive and positional information. While

DNABERT focuses on contextual features, LSTM preserves the sequential structure of genomic data, improving mutation classification.

LSTM uses memory cells and gating mechanisms (input, forget, output gates) to control information flow and overcome the vanishing gradient problem. This allows it to retain important information over long sequences and detect long-distance interactions and subtle mutation patterns in DNA sequences.

3.11 Evaluation Metrics

Accuracy refers to the ratio of the number of correct predictions made by a model to the total number of predictions. Mathematically, This is determined by taking the ratio of the sum of true positives and true negatives to the total number of instances. In genomic sequence classification with DNABERT, accuracy means the model's ability of correctly classifying sequences into different categories, like types of mutations, regulatory elements, or pathogenic variants. While accuracy is an unambiguous performance metric, it could be argued that it is not the best measure for an imbalanced dataset where one class is inappropriately larger, for the reason that the model could obtain high accuracy based on just predicting the majority class.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

F1 score is the harmonic mean of precision and recall which balances the two metrics, presenting a more inclusive picture of performance, especially in cases of unbalanced class distributions. The F1 score is particularly relevant in genomic classification tasks with DNABERT, especially when rare mutations or regulatory sequences need to be detected. In this regard, a high F1 score indicates that the model successfully identifies positive instances while generating low rates of false positives and false negatives and hence has a good and trusted classification.

$$\mathbf{F1\ Score} = \frac{2 \times \mathit{Precision} \times \mathit{Recall}}{\mathit{Precision} + \mathit{Recall}}$$

3.12 Feasibility Study

3.12.1 Economic Feasibility

The proposed system demonstrates strong economic feasibility with minimal infrastructure requirements and negligible operational costs. The system operates using pretrained models without requiring expensive GPU resources for training or fine-tuning. Standard CPU-based hardware is sufficient for deployment, with a recommended minimum of 4GB RAM and a dual-core processor capable of running inference in under 100 milliseconds per evaluation.

Development costs are primarily limited to one-time implementation effort, as the system leverages freely available open-source libraries including sentence-transformers, spaCy, and Streamlit. No licensing fees or recurring API costs are incurred. The lightweight model size (80MB) enables deployment on modest hardware without expensive cloud computing infrastructure. Educational institutions can deploy the system on existing computer lab infrastructure or standard classroom hardware. For online deployment, basic web hosting supporting Python applications suffices, with estimated monthly costs under \$10-20 for serving hundreds of students. Compared to the ongoing personnel costs of manual grading, the system provides substantial long-term cost savings while maintaining consistent grading quality.

3.12.2 Operational Feasibility

Operational feasibility is high due to the system's user-friendly interfaces and straightforward deployment process. Both commandline and web-based interfaces require minimal training for instructors and students to use effectively. The question addition process involves simply updating the JSON reference bank file without technical modifications to the codebase, enabling nontechnical educators to expand the question set.

The system integrates naturally into existing educational workflows, supporting both formative assessment during learning sessions and summative assessment in formal examinations. The real-time evaluation capability enables immediate feedback, enhancing the learning experience compared to delayed manual grading. Multiple deployment modes (standalone application, web service, or integration into learning management systems) provide flexibility to accommodate diverse institutional requirements.

Maintenance requirements are minimal, primarily involving periodic updates to the reference bank with new questions and reference answers. The system operates autonomously without requiring ongoing supervision or intervention during normal operation. Error handling mechanisms provide clear messages when issues occur, facilitating troubleshooting.

3.12.3 Technical Feasibility

Technical feasibility is well-established through the use of mature, well-documented open-source technologies with extensive community support. The sentence-transformers library provides robust, production-ready implementations of state-of-the-art embedding models. The spaCy library offers reliable natural language processing functionality with comprehensive documentation and proven performance in educational applications.

The system architecture follows established software engineering principles including modularity, separation of concerns, and clear interfaces between components. The codebase is organized into logical modules that can be independently tested, updated, and extended. Standard Python development tools and practices ensure maintainability and facilitate future enhancements.

Deployment technical requirements are readily achievable: Python 3.7 or higher, standard scientific computing libraries (numpy, pandas), and approximately 200MB of disk space for models and dependencies. The system supports both Windows and Linux operating systems without platform-specific modifications. Installation follows standard Python package management practices using pip, with a documented setup process requiring minimal technical expertise.

Performance testing demonstrates the system meets real-time evaluation requirements, with average response times of 50-80 milliseconds per answer evaluation on standard hardware. The architecture supports concurrent evaluation of multiple student answers through parallel processing, enabling scalability to classroom-sized deployments without performance degradation.

Chapter 4

System Requirements

4 SYSTEM REQUIREMENTS

4.1 Functional Requirements

The proposed system for genomic mutation classification using DNABERT-2 and LSTM is designed to perform a sequence of operations that enable accurate and efficient analysis of DNA data. The system should be capable of accepting genomic sequences from reliable sources such as the NCBI Viruses database and loading them for processing. It must preprocess the data by cleaning invalid or noisy sequences, removing inconsistencies, and standardizing the input format [19]. After preprocessing, the system should convert DNA sequences into k-mer representations, followed by tokenization and encoding using the AutoTokenizer, generating input IDs and attention masks suitable for transformer-based models.

The system should then utilize the pretrained DNABERT-2 model to extract contextual embeddings from the tokenized sequences. These embeddings must be further processed using an LSTM layer to capture long-range dependencies and sequential patterns in genomic data. Based on these learned features, the system should classify sequences into categories such as wildtype or mutated using a fully connected output layer. Additionally, the system must support model training with labeled data, including optimization and parameter tuning [17], [21]. It should also evaluate performance using metrics such as accuracy, precision, recall, and F1-score. Finally, the system should provide predictions for new input sequences and present the results in a clear and interpretable manner, ensuring usability for genomic analysis tasks [23].

4.2 Non-Functional Requirements

The proposed genomic mutation classification system using DNABERT-2 and LSTM must satisfy several non-functional requirements to ensure efficiency, reliability, and usability. The system should provide high performance by processing large-scale genomic datasets efficiently, with optimized memory usage and faster computation through techniques such as batch processing and GPU acceleration. It must ensure scalability, allowing it to handle increasing volumes of genomic data without significant degradation in performance.

The system should maintain accuracy and reliability, producing consistent and correct predictions across different datasets. It must also ensure robustness, handling noisy, incomplete, or variable-length DNA sequences without failure. Usability is important, so the system should present outputs in a clear and interpretable format, making it easier for researchers to understand the results [6], [22].

Additionally, the system should support maintainability and flexibility, allowing easy updates, model improvements, and integration with new datasets or tools. It must ensure security and data integrity, protecting sensitive genomic data from unauthorized access or corruption. Finally, the system should provide portability, enabling deployment across different platforms and environments, such as local machines or cloud-based systems, ensuring accessibility for various research applications [1], [9].

4.2.1 Software Requirements

Table 4.1: Software Requirements

Category	Technology / Tool	Version
Programming Language	Python	3.8 or above
Data Handling	Pandas	1.3+
Numerical Computing	NumPy	1.21+
Deep Learning Framework	PyTorch	1.10+
NLP Library	Transformers (Hugging Face)	4.20+
Visualization	Matplotlib / Seaborn	3.5+ / 0.11+
Development Environment	Jupyter Notebook / Google Colab	Latest

The proposed system for genomic mutation classification using DNABERT-2 and LSTM is implemented using the Python programming language, which provides strong support for machine learning and bioinformatics applications. Key libraries include Pandas and NumPy for data handling and numerical computations. The Hugging Face Transformers library is used to implement DNABERT-2 and AutoTokenizer for sequence processing, while PyTorch is used for building and training the deep learning model, including LSTM layers.

For visualization, tools such as Matplotlib or Seaborn are used to plot accuracy and loss graphs. The system is developed in environments like Jupyter Notebook, Google Colab, or Anaconda, which support interactive coding and experimentation. The software is compatible with Windows, Linux, and macOS, and supports GPU acceleration (CUDA) for faster training and efficient processing of large genomic datasets.

4.2.2 Hardware Requirements

Table 4.2: Hardware Requirements

Component	Minimum Requirement	Recommended Requirement
Processor	Intel i5 / AMD Ryzen 5	Intel i7 / AMD Ryzen 7 or above
RAM	8 GB	16 GB or higher
Storage	256 GB SSD	512 GB SSD or higher
GPU	Not mandatory (CPU supported)	NVIDIA GPU with CUDA support
System Type	64-bit system	64-bit system
Network	Basic internet connection	High-speed internet connection

The proposed system requires a standard computing setup to efficiently process genomic data and train the DNABERT-2 + LSTM model. A system with a minimum of an Intel i5 or AMD Ryzen 5 processor is recommended to handle data processing tasks effectively. The system should have at least 8 GB of RAM, although 16 GB is preferred for better performance when working with large genomic datasets.

In terms of storage, a minimum of 256 GB SSD is required to store datasets, trained models, and outputs, with higher capacity recommended for large-scale analysis. For faster model training and improved computational efficiency, the use of an NVIDIA GPU with CUDA support is highly recommended, although the system can also run on CPU with slower performance. The system should be a 64-bit architecture, ensuring compatibility with modern machine learning frameworks and libraries. Additionally, a stable internet connection is necessary for downloading datasets, libraries, and updates. Overall, these hardware requirements ensure smooth execution, scalability, and efficient performance of the proposed genomic mutation classification system.

Chapter 5

Design

5 DESIGN

5.1 System design

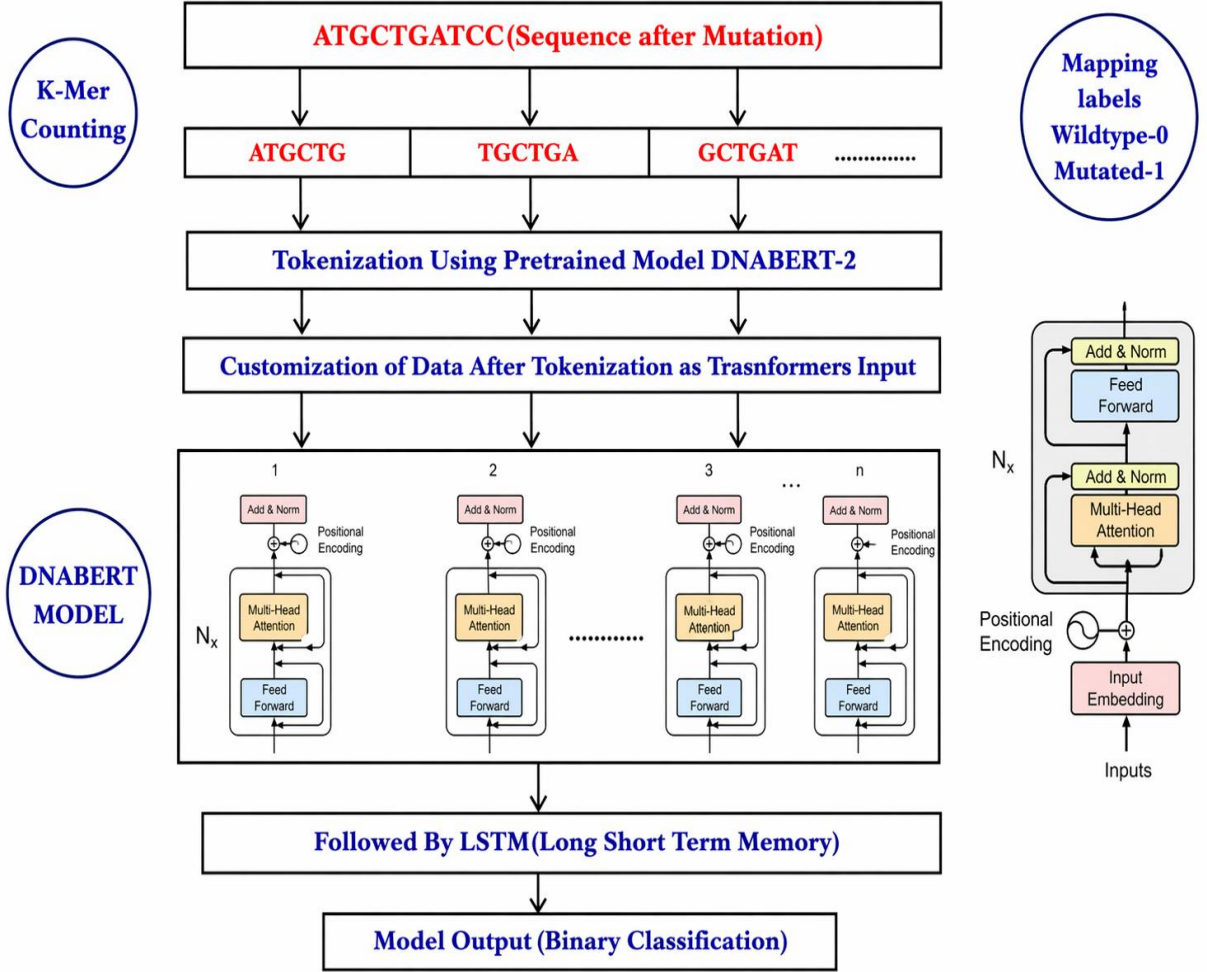


Figure 5.1: System Architecture

DNABERT analyzes these tokenized sequences using a self-attention mechanism, which allows the model to focus on important parts of the sequence while capturing contextual relationships between nucleotides [10], [14]. This mechanism helps in identifying mutation-relevant regions and reduces the impact of noise in the data. The model generates high-dimensional contextual embeddings that encode both local features and bidirectional dependencies across the DNA sequence. These embeddings serve as a rich representation of genomic information and are used as input for further sequential analysis [18].

To enhance long-range pattern recognition, an LSTM layer is added after the DNABERT model. While DNABERT captures contextual relationships, LSTM networks are specifically designed to maintain sequential dependencies over longer stretches of data through memory cells and gating mechanisms [2], [10]. This combination ensures that both short-term and long-term dependencies in genomic sequences are effectively captured. The LSTM processes the embeddings sequentially and learns complex mutation patterns that may span across distant regions of the DNA sequence.

The final output from the LSTM layer is passed to a fully connected dense layer for binary classification. A sigmoid activation function is used to generate a probability score indicating whether a sequence is wild-type or mutated. The model is trained using optimization techniques such as backpropagation and binary cross-entropy loss, ensuring accurate classification performance.

By combining transformer-based contextual learning with sequential modeling, the DNABERT-LSTM architecture significantly improves mutation detection accuracy and robustness. This hybrid framework effectively captures both contextual and sequential features, making it highly suitable for viral genome analysis, mutation tracking, and large-scale genomic data processing [1], [4], [22].

5.2 Data Flow of System

The data flow of the proposed DNABERT-2 + LSTM mutation classification system follows a structured pipeline from input to output. Initially, genomic DNA sequences are collected from sources such as the NCBI Viruses dataset. These raw sequences are passed to the data preprocessing stage, where noise, invalid characters, and inconsistencies are removed, and the data is standardized for further processing.

After preprocessing, the sequences are transformed into k-mer representations, which divide the DNA sequence into overlapping substrings. These k-mers are then processed using the AutoTokenizer, which converts them into numerical formats such as input IDs and attention masks suitable for transformer-based models.

The tokenized data is then fed into the DNABERT-2 model, which performs feature extraction by generating contextual embeddings using self-attention mechanisms. These embeddings capture relationships between nucleotides within the sequence. The output from DNABERT-2 is passed to the LSTM layer, which models sequential dependencies and long-range patterns in the genomic data.

Finally, the processed features are sent to a fully connected classification layer, which predicts the class label—either wildtype or mutated. The result is then presented as the final output of the system. This structured data flow ensures efficient processing, accurate feature learning, and reliable mutation classification.

Chapter 6

Implementation

6 IMPLEMENTATION

6.1 Implementation Overview

BEGIN

1. Mount Google Drive
2. Install required libraries (transformers, datasets)
3. Load dataset from CSV file
4. Extract DNA sequences and mutation labels
5. FOR each DNA sequence IN dataset DO

 Convert sequence into k-mers ($k = 6$)

END FOR

6. Map labels:

 IF mutation_class == "wildtype" THEN label = 0

 ELSE IF mutation_class == "mutated" THEN

label = 1

END IF

7. Remove rows with missing labels
 8. Load DNABERT-2 tokenizer
 9. FOR each k-mer sequence DO
- Tokenize sequence
- Apply padding and truncation ($\text{max_length} = 512$)
- Generate input_ids and attention_mask
- END FOR

10. Split dataset into Training set (80%) and
Validation set (20%)

11. Create custom Dataset class
12. Create DataLoader for training and validation
13. Load pre-trained DNABERT model
14. Initialize BERT-LSTM model:
 - DNABERT embedding layer
 - Bidirectional LSTM layer
 - Fully connected layers
15. Move model to GPU (if available)
16. Compute class weights
17. Define BCEWithLogitsLoss
18. Initialize AdamW optimizer
19. Initialize learning rate scheduler
20. Set early stopping parameters
21. FOR each epoch DO
 - Set model to training mode
 - FOR each batch in training data DO
 - Forward pass through DNABERT
 - Pass embeddings to LSTM
 - Compute predictions
 - Calculate loss
 - Backpropagate gradients
 - Clip gradients
 - Update model parameters
- END FOR

```

    Evaluate model on validation set

    Compute validation loss, accuracy, F1-score,
ROC-AUC

    IF validation loss improves THEN

        Save model

        Reset patience counter

    ELSE

        Increase patience counter

    END IF

    IF patience counter >= limit THEN

        Stop training (Early Stopping)

    END IF

END FOR

22. Load best saved model

23. Perform final evaluation

24. Output Accuracy, F1-score, ROC-AUC

END

```

The implementation of the proposed genomic mutation classification system begins by mounting Google Drive and loading the dataset from a CSV file. The DNA sequences and corresponding mutation labels are extracted, and each sequence is converted into k-mer representations ($k = 6$) to make the data suitable for NLP-based processing. The mutation labels are then encoded into binary values, where wildtype is mapped to 0 and mutated to 1. Any rows with missing labels are removed to ensure data quality.

Next, the DNABERT tokenizer is loaded to process the k-mer sequences. Each sequence is tokenized with appropriate padding and truncation, generating input IDs and attention

masks required for transformer-based models. The dataset is then split into training (80%) and validation (20%) sets, maintaining class balance. A custom dataset class and DataLoader are created to efficiently handle batch processing during training.

The pretrained DNABERT model is initialized, and a hybrid BERT-LSTM architecture is constructed by combining DNABERT embeddings with a bidirectional LSTM layer, followed by fully connected layers for classification. The model is moved to GPU if available for faster computation. To handle class imbalance, class weights are calculated and used in the Binary Cross-Entropy with Logits loss function. The training process is configured using the AdamW optimizer, learning rate scheduler, and early stopping mechanism.

During training, the model performs forward propagation through DNABERT and LSTM layers, computes loss, and updates weights using backpropagation. Gradient clipping is applied to ensure stable training. After each epoch, the model is evaluated on the validation set using metrics such as validation loss, accuracy, F1-score, and ROC-AUC. The best model is saved based on improved validation performance, and early stopping is triggered if no improvement is observed.

Finally, the best saved model is loaded for final evaluation, and the performance metrics are reported. This structured approach ensures efficient training, improved accuracy, and reliable mutation classification.

Chapter 7

Results

7 RESULTS

7.1 Result Flow

The results flow of the proposed DNABERT-2 + LSTM model describes how the system produces final outputs after training and evaluation. Initially, the trained model receives validation or test genomic sequences as input. These sequences undergo the same preprocessing steps, including k-mer conversion and tokenization, to ensure consistency with the training phase. The processed inputs are then passed through the DNABERT-2 model, which generates contextual embeddings, followed by the LSTM layer that captures sequential dependencies [9], [17].

The model then produces predictions through the final classification layer, where each sequence is classified as either wildtype or mutated. These predictions are compared with the actual labels to evaluate performance [2], [6].

Next, the system computes evaluation metrics such as accuracy, precision, recall, F1-score, and ROC-AUC. These metrics help measure the effectiveness of the model in correctly identifying mutation patterns. The results are also visualized using graphs such as training vs validation loss and accuracy curves, providing insight into model performance and convergence.

Finally, the best-performing model (saved during training) is used for final evaluation and prediction on new data. The output is presented in a clear format, including predicted labels and performance scores [5]. This results flow ensures that the system provides accurate, reliable, and interpretable outputs for genomic mutation classification.

7.2 Training Loss Analysis

The variation of training loss across different epochs during the learning process of the proposed model. The graph shows a clear decreasing trend in the loss value as the number of epochs increases, indicating that the model is progressively learning meaningful patterns from the genomic sequence data. At the initial stage, the loss value is relatively high (close to 1.0), which suggests that the model predictions are not yet aligned with the actual labels.

As training proceeds, there is a sharp decline in loss from the first to the third epoch, reflecting rapid learning and effective adjustment of model parameters. This phase indicates that the model is quickly capturing important features from the input data. Between the third and fourth epochs, the loss continues to decrease but at a slower rate, showing that the learning process is stabilizing.

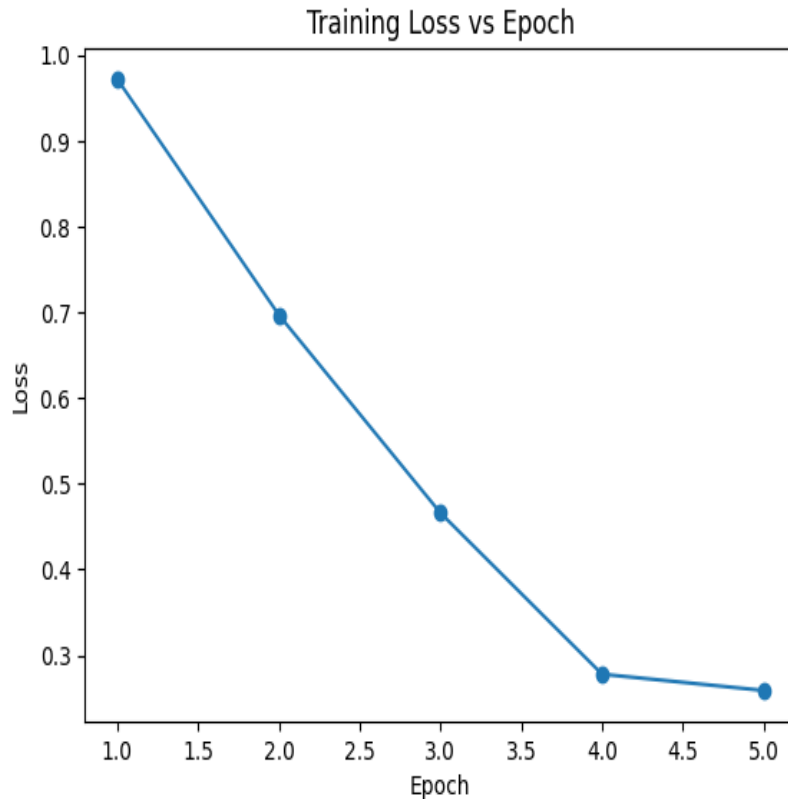


Figure 7.1: Training Loss Analysis

In the final epochs, the loss curve becomes relatively flat, reaching a value close to 0.25, which indicates that the model has approached convergence. The stabilization of loss suggests that further training would result in only marginal improvements. Overall, the consistent reduction in training loss demonstrates that the proposed DNABERT-2 and LSTM based model is effectively optimizing its parameters and improving prediction accuracy over time.

7.3 Training Accuracy Analysis

The training accuracy versus epoch during the training process of the proposed model. The graph shows a consistent increase in accuracy as the number of epochs increases, indicating that the model is effectively learning patterns from the genomic sequence data. At the initial stage, the model achieves an accuracy of approximately 66%, which reflects the early learning phase where the model begins to understand basic sequence features.

As training progresses, the accuracy improves steadily, reaching around 70% in the second epoch and 75% in the third epoch, demonstrating that the model is capturing meaningful relationships within the DNA sequences. This gradual improvement highlights the effectiveness of the feature extraction and sequential learning process.

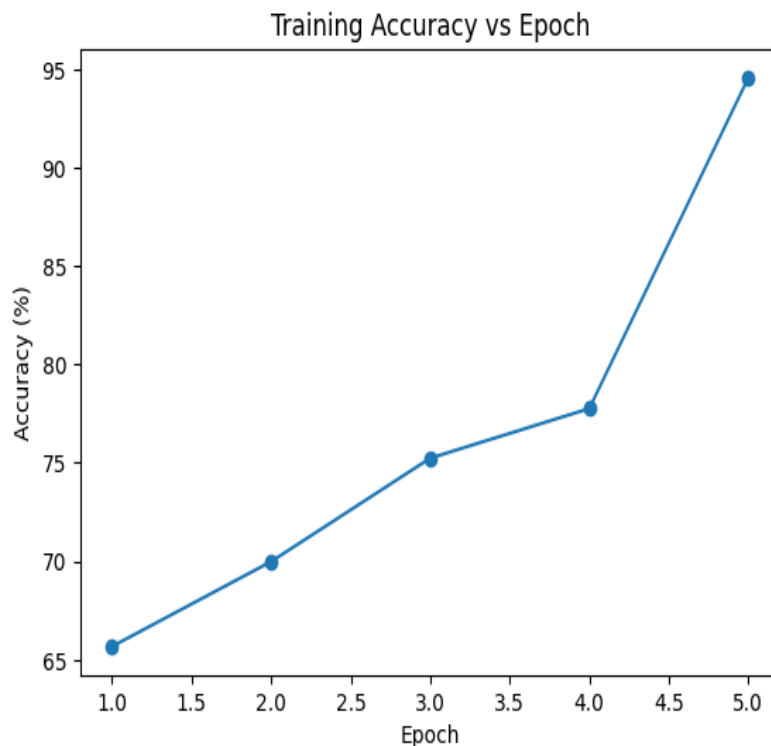


Figure 7.2: Training Accuracy Analysis

Between the third and fourth epochs, the accuracy continues to increase but at a moderate rate, indicating that the model is refining its learned representations. A significant improvement is observed in the final epoch, where the accuracy rises sharply to approximately 94–95%, showing that the model has successfully learned both contextual and long-range dependencies in the data.

The overall upward trend of the accuracy curve indicates stable and efficient training. The high final accuracy confirms that the proposed DNABERT-2 and LSTM based model is capable of accurately classifying genomic sequences and detecting mutations with strong performance.

7.4 Comparison of Models

Comparison between Convolutional Neural Network (CNN) and Long Short-Term Memory (LSTM) models for genomic sequence classification. The graph clearly illustrates the difference in performance between the two models in terms of classification accuracy. The CNN model achieves an accuracy of approximately 76%, while the LSTM model attains a significantly higher accuracy of around 93%.

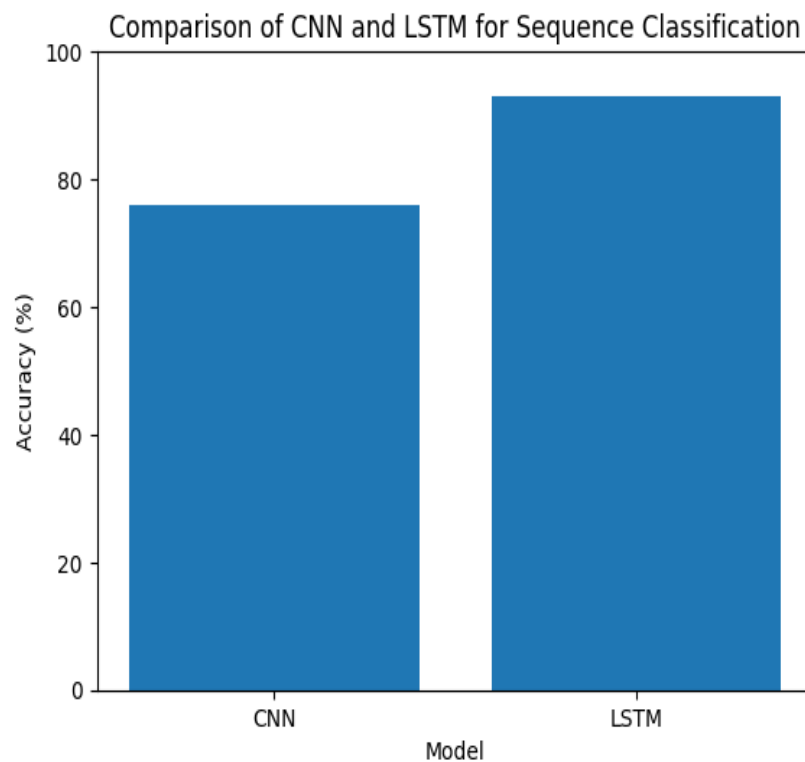


Figure 7.3: Comparison of Models

CNN models are primarily designed to extract local features and spatial patterns from input data using convolution operations. While they are effective in identifying short-range patterns in genomic sequences, they have limitations in capturing long-range dependencies across distant nucleotide positions. This makes them less suitable for complex biological

sequences where interactions may occur over extended regions.

In contrast, LSTM networks are specifically designed for sequential data processing and are capable of maintaining long-term dependencies through memory cells and gating mechanisms. This allows the LSTM model to effectively learn relationships between distant elements in DNA sequences, which is crucial for accurate mutation classification.

The results demonstrate that the LSTM model significantly outperforms the CNN model due to its ability to capture both temporal and sequential dependencies in genomic data. This highlights the importance of using sequence-aware models for biological data analysis. Overall, the comparison confirms that LSTM-based architectures are more suitable for genomic sequence classification tasks, especially when long-range relationships play a critical role.

7.5 Results

The proposed DNABERT-2 + LSTM model was evaluated on Human Virus, Influenza Virus, and SARS-CoV-2 datasets. The model achieved high performance across all datasets.

Table 7.1: Experimental Results

Data set	Accuracy (%)	F1-Score (%)
Human Virus	93.4	92.0
Influenza Virus	90.7	89.0
SARS-CoV-2	92.3	90.0

It obtained 93.4% accuracy and 92.0% F1-score for Human Virus, 90.7% accuracy and 89.0% F1-score for Influenza Virus, and 92.3% accuracy and 90.0% F1-score for SARS-CoV-2. Overall the model shows accurate and reliable mutation classification across different genomic datasets.

Chapter 8

Social Impact

8 SOCIAL IMPACT

The proposed DNABERT-2 and LSTM based mutation classification system has a wide-ranging social impact across healthcare, public health, and genomic research. By enabling accurate identification of mutations in viral genomes, the system supports early detection, monitoring, and surveillance of infectious diseases. This allows health authorities to track emerging variants, respond quickly to outbreaks, and implement timely preventive measures, thereby strengthening overall public health preparedness and reducing the spread of diseases [1], [22].

One of the major contributions of this system is in pandemic management and control. By efficiently analyzing mutation patterns in viruses such as SARS-CoV-2, the model helps researchers understand how viruses evolve, spread, and adapt [6], [7]. This information is crucial for identifying new variants, assessing their transmissibility, and planning appropriate containment strategies, ultimately supporting global efforts in managing future pandemics.

The system also significantly contributes to precision medicine, where treatments are tailored based on an individual's genetic makeup. Accurate mutation classification enables better diagnosis of genetic disorders and supports the development of personalized therapies, improving treatment effectiveness and patient outcomes. In addition, the system aids in drug discovery and vaccine development by providing insights into mutation patterns, helping scientists design more effective vaccines and targeted drugs, especially for rapidly mutating viruses [17], [21].

From a technological perspective, the integration of artificial intelligence and deep learning in genomics reduces dependency on manual analysis, making genomic research faster, scalable, and more efficient. This not only accelerates scientific discoveries but also promotes interdisciplinary collaboration between computer science, biology, and healthcare. It opens new opportunities for innovation in bioinformatics and encourages students and researchers to explore advanced AI-driven solutions in life sciences.

Furthermore, the system improves accessibility to advanced diagnostic tools, especially in resource-limited or developing regions. By providing automated and cost-effective solutions, it helps bridge the gap between advanced research and real-world healthcare applications. This can lead to better healthcare services, early disease detection, and improved health outcomes for a larger population [4], [22].

Overall, the proposed system contributes to global health improvement, technological advancement, and scientific progress, making it a valuable tool in modern healthcare and biomedical research.

Chapter 9

Conclusion & Future Work

9 CONCLUSION & FUTURE WORK

9.1 Conclusion

In this study, a hybrid deep learning framework combining DNABERT-2 and Long Short-Term Memory (LSTM) networks was proposed for the classification of mutations in viral genomic sequences. The model effectively integrates the contextual learning capability of transformer-based architectures with the sequential modeling strength of recurrent neural networks. By utilizing k-mer tokenization and DNABERT-2 embeddings, the system captures meaningful contextual relationships within DNA sequences, while the LSTM layer models long-range dependencies that are essential for accurate genomic analysis.

The experimental evaluation demonstrates that the proposed model achieves high accuracy and F1-score across multiple viral datasets, including Human Virus, Influenza Virus, and SARS-CoV-2. The analysis of training loss and accuracy curves indicates stable learning behavior and successful convergence of the model. The steady decrease in loss and consistent increase in accuracy confirm that the hybrid architecture effectively learns complex genomic patterns. Furthermore, the comparison with traditional models such as CNN highlights the advantage of LSTM-based approaches in handling sequential and long-range dependencies in DNA sequences [10], [14], [18].

The improved performance of the proposed model can be attributed to the complementary strengths of DNABERT-2 and LSTM. While DNABERT-2 captures bidirectional contextual information using attention mechanisms, LSTM enhances the model's ability to retain sequential information over long genomic regions. This combination enables the model to identify subtle mutation patterns that may not be captured by standalone models. As a result, the hybrid framework provides a more robust and reliable solution for genomic mutation classification [2], [6].

In addition to performance improvements, the proposed approach offers scalability and adaptability for large-scale genomic data analysis. The use of pretrained transformer models reduces the need for extensive manual feature engineering and allows efficient

processing of complex biological sequences. The framework can also be extended to other genomic applications such as gene prediction, disease mutation identification, and regulatory element detection [9], [14].

Overall, this research highlights the potential of NLP-based deep learning techniques in transforming genomic data analysis. The proposed DNABERT-2 and LSTM hybrid model provides an effective, scalable, and accurate solution for mutation classification and demonstrates the growing importance of artificial intelligence in bioinformatics [1], [9]. This work contributes to advancing computational genomics and supports future developments in automated genomic analysis and precision medicine.

9.2 Future Work

Although the proposed DNABERT-2 and LSTM hybrid model demonstrates promising performance in genomic mutation classification, several opportunities exist to further enhance its effectiveness and applicability.

One important direction for future work is the utilization of larger, more diverse, and high-quality genomic datasets. Expanding the dataset to include a wider range of viral genomes and mutation annotations from different species can significantly improve the robustness and generalization capability of the model. Incorporating data from large-scale genomic repositories may help the model learn complex mutation patterns across diverse biological conditions [3], [7].

Another key extension is the development of multi-class and fine-grained mutation classification models. The current approach focuses on binary classification (wildtype vs mutated), but future research can extend this framework to identify specific mutation types such as single nucleotide polymorphisms (SNPs), insertions, deletions, and structural variations. This would provide more detailed biological insights and enhance the practical utility of the model in genomic studies.

Future work can also explore advanced transformer architectures and large-scale genomic foundation models. Incorporating more recent transformer variants or pretraining strategies

tailored for biological data may improve contextual representation and overall classification performance. Fine-tuning domain-specific pretrained models on genomic datasets can further enhance accuracy and efficiency [5], [9].

Another promising direction is the integration of multi-modal biological data, such as gene expression profiles, epigenetic information, protein structures, and regulatory elements. Combining sequence-based features with additional biological data sources can provide a more comprehensive understanding of mutation effects and improve predictive performance.

In addition, improving the interpretability and explainability of the model is an important area for future research. Techniques such as attention visualization, feature importance analysis, and explainable AI methods can help identify which regions of the DNA sequence contribute most to mutation predictions, making the model more transparent and useful for biological researchers.

Future research can also focus on developing real-time and scalable genomic analysis systems. Such systems could enable rapid detection and classification of mutations during viral outbreaks, supporting applications in disease monitoring, vaccine development, and public health response [13], [17].

Another potential enhancement is optimizing the model for computational efficiency and deployment. Reducing model complexity, improving inference speed, and enabling deployment on edge or cloud-based platforms can make the system more practical for real-world applications, including clinical and research environments.

Finally, the proposed framework can be extended to other important applications such as gene prediction, disease diagnosis, drug discovery, and evolutionary analysis. By adapting the model to different bioinformatics tasks, the impact of NLP-based deep learning techniques in genomics can be further expanded.

Chapter 10

Bibliography

10. BIBLIOGRAPHY

- [1] Ruan, W., Lyu, Y., Zhang, J., Cai, J., Shu, P., Ge, Y., ... & Liu, T. (2025). Large language models for bioinformatics. *Quantitative Biology*, 14(1), e70014.
- [2] Anjana, M. P., & Arun, K. S. (2025). Transformer-based models for uncovering genetic mutations in cancerous and non-cancerous genomes. *Gene*, 963, 149460.
- [3] Huttunen, A. (2025). Multimodal Tumour Type and Subtype Classification with Deep Learning.
- [4] Robertson, J., Consul, S., & Vikalo, H. (2025). NextVir: Enabling classification of tumor-causing viruses with genomic foundation models. *PLOS Computational Biology*, 21(8), e1013360.
- [5] Marathe, V. (2025). Exploring and Predicting Viral Evolution Using Large Language Models (Doctoral dissertation, North Dakota State University).
- [6] Hegde, M., Nebel, J. C., & Rahman, F. (2025). Language modelling techniques for analysing the impact of human genetic variation. *Bioinformatics and Biology Insights*, 19, 11779322251358314.
- [7] Marathe, V., Bajracharya, D., & Yan, C. (2025, November). VirusT5: harnessing large language models to predicting SARS-CoV-2 evolution. In 2025 IEEE 25th International Conference on Bioinformatics and Bioengineering (BIBE) (pp. 676–681). IEEE.
- [8] Gunasekaran, H., Wilfred Blessing & Husain (2025). Enhanced viral genome classification using large language models. *Algorithms*, 18(6), 302.
- [9] Ruan, W., Lyu, Y., Zhang, J., Cai, J., Shu, P., Ge, Y., ... & Liu, T. (2025). Large language models for bioinformatics. *Quantitative Biology*, 14(1), e70014.
- [10] Kabir, A., Bhattarai, M., Peterson, S., Najman-Licht, Y., Rasmussen, K. Ø., Shehu, A., ... & Usheva, A. (2024). DNA breathing integration with deep learning foundational model advances genome-wide binding prediction of human transcription factors. *Nucleic Acids Research*, 52(19), e91–e91.
- [11] Schoier, A. F. (2024). Taxonomic inference with DNA foundation models.
- [12] Attique, M., Khan, Y. D., Alturise, F., & Alkhalifah, T. (2024). Deep learning based prediction of RNA 5hmC modifications using composite feature representations and comparative benchmarking with transformer models. *BioData Mining*, 19, 15.

- [13] Piroozeh, Z., Bazarova, A., & Kalinina, O. V. (2024). Unravelling DNA Base Composition Behind Budding Yeast DNA Replication Origins Using Large Language Models (No. FZJ-2026-00950). Jülich Supercomputing Center.
- [14] Sun, J., Huang, Y., Fu, J., Teng, L., Liu, X., & Luo, X. (2024). An enhanced BERT model with improved local feature extraction and long-range dependency capture in promoter prediction for hearing loss. *PeerJ Computer Science*, 11, e3104.
- [15] Wang, F., Xia, Y., Jia, Y., Zhang, Z., Deng, Y., Wu, Y., & Zhang, Y. (2024). *AI Med.*
- [16] Dip, S. A., Shuvo, U. A., Chau, T., Song, H., Choi, P., Wang, X., & Zhang, L. (2024). PathoLM: Identifying pathogenicity from the DNA sequence through the Genome Foundation Model. *arXiv preprint arXiv:2406.13133*.
- [17] Almuhaideb, S., Altwaijry, N., Al-Turaiki, I., Khan, A. R., & Rizvi, H. A. (2023). A comparative study of data representation techniques for deep learning-based classification of promoter and histone-associated DNA regions. *Computers, Materials, & Continua*, 85(2), 3095.
- [18] Testagrose, C., & Boucher, C. (2023). Tokenization and deep learning architectures in genomics: A comprehensive review. *Computational and Structural Biotechnology Journal*.
- [19] Zhang, M., Wu, J., Wang, Y., Cao, Y., Liu, J., Wang, Q., ... & Wang, Y. (2023). Deep-m7G: A contrastive learning-based deep biological language model for identifying RNA N7-methylguanosine sites. *International Journal of Biological Macromolecules*, 318, 145341.
- [20] Gautreau, G. (2023). What can genomics learn from natural language processing? (Doctoral dissertation, Université paris Cité).
- [21] Zhou, Z. (2023). Deciphering the Language of DNA with Genome Foundation Models (Doctoral dissertation, Northwestern University).
- [22] Galeeva, J., Kuzmichenko, P., Manolov, A., Lukashev, A., & Ilina, E. (2023). Bioinformatics Tools and Approaches for Virus Discovery in Genomic Data: A Systematic Review. *Viruses*, 17(12), 1538.
- [23] Ligeti, B., Szepesi-Nagy, I., Bodnár, B., Ligeti-Nagy, N., & Juhász, J. (2022). ProkBERT family: genomic language models for microbiome applications. *Frontiers in Microbiology*, 14, 13.